

Divit Rawal

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Research Interests: statistical/machine learning theory; generalization and implicit bias in overparameterized models; geometric statistics and information geometry; stochastic processes and statistical physics of learning.

EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Statistics, Physics, Computer Science

2023 – 2027

Relevant Coursework: Statistical Learning Theory[†]; Theoretical/Mathematical Statistics[†]; Convex Optimization; Learning for Dynamics and Control[†]; Probability and Random Processes; Statistical/Thermal Physics; Randomized Linear Algebra, Optimization, Large-Scale Learning[†]; Statistical Computing. ([†] denotes graduate-level coursework)

Teaching: Head TA (Sp26), Reader (Sp25) for Communication Networks (EE 122). Developed lab curriculum, wrote and graded exams, and led weekly recitation sessions.

PUBLICATIONS AND PREPRINTS

A Theory of Saddle Escape in Deep Nonlinear Networks

2026

Divit Rawal, Michael R. DeWeese

Under review, NeurIPS.

[\[arXiv\]](#)

Rao-Blackwellized Score Matching on Manifolds

2026

Divit Rawal

Under review, ICML Workshop on Structured Probabilistic Inference and Generative Modeling.

Minimax Rates for Hyperbolic Hierarchical Learning

2026

Divit Rawal, Sriram Vishwanath

Preprint.

[\[arXiv\]](#)

ALPHANSO: Open-Source (α, n) Neutron Source Terms

2026

Divit Rawal, Anthony J. Nelson, William Zywiec, Daniel Siefman

2026 American Nuclear Society Student Conference.

[\[arXiv\]](#)

2026 Institute of Nuclear Materials Management Annual Meeting.

[\[GitHub\]](#)

Under review, Nuclear Instruments and Methods in Physics Research Section A.

AWARDS AND GRANTS

Best Paper Award (ANS Student Conference 2026)

Apr 2026

- Best Paper in Mathematics, Computation, and AI Applications at the American Nuclear Society Student Conference for “ALPHANSO: Open-Source Modeling of (α, n) Source Terms,” selected from ~ 250 submissions.

VESSL AI Academia Grant

Aug 2025

- Awarded academic grant for independent research into embeddings and information geometry; sole undergraduate awardee among recipients primarily comprising Ph.D. students and postdoctoral researchers.

RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory

Mar 2025 – Present

Researcher (Nuclear Science and Security Consortium Fellow)

Livermore, CA

- Developed [ALPHANSO](#), an open-source computational framework for deterministic modeling of (α, n) neutron production using continuous slowing-down approximations and modern evaluated nuclear data.
- Formulated and implemented stochastic and deterministic transport models for neutron yield and spectral prediction, benchmarking against legacy codes and experimental datasets.
- Conducted quantitative validation against experimental neutron yield and spectral benchmarks, analyzing model error and sensitivity to nuclear cross-section data; demonstrated improved predictive accuracy and extensibility relative to legacy SOURCES-based pipelines.

Berkeley Artificial Intelligence Research (BAIR)

Sep 2024 – Present

Researcher, DeWeese Lab

Berkeley, CA

- Authored paper deriving an exact layer-imbalance identity for deep nonlinear networks and a critical-depth escape law, with matching theory and simulation under both symmetric and He-normal initialization.
- Analyzed in-context learning in kernel ridge regression via eigenlearning, deriving mode-wise error dynamics and conditions for generalization without representation change.
- Connected ICL behavior to spectral bias in kernel methods (eigenvalue decay, target alignment), clarifying how contextual examples induce effective adaptation in fixed-feature models.

UC Irvine, Department of Physics & Astronomy

Feb 2022 – Jul 2023

Researcher, Whiteson Lab

Irvine, CA

- Developed deep learning models using TensorFlow/Keras for high-energy particle collision analysis, achieving 90% accuracy (vs. 80% baseline) in particle identification.
- Simulated particle collisions using MadGraph, Pythia8, Delphes, and ROOT; implemented reconstruction algorithms in C++ and Python achieving 2% mass prediction error.
- Contributed to [arXiv:2412.01600](#) and [arXiv:2412.01548](#), both published in *Reports on Progress in Physics*.

INDUSTRY EXPERIENCE

Cisco Systems (Foundation AI)

Jun 2025 – Aug 2025

Software Engineer (Intern)

San Francisco, CA

- Contributed to post-training and evaluation of cybersecurity-specialized LLMs built on Llama 3.1 8B, including reasoning, instruction tuning, and preference alignment (RLHF) for security dialogue and workflows.
- Supported open-weight releases [Foundation-Sec-8B](#) and [Foundation-Sec-8B-Instruct](#), and contributed to the accompanying technical reports documenting data curation, training methodology, and benchmark evaluation.

ExperienceFlow AI

May 2024 – Sep 2024

Machine Learning Engineer (Intern)

Remote

- Modeled finite state machine (FSM) transition dynamics using transformer, state-space, and graph architectures, evaluating generalization and sample efficiency in low-label sequence prediction settings.
- Developed reinforcement-learning-based control policies (Deep Q-Learning, SARSA) for dynamic state reasoning systems, improving stability and convergence in sequential decision tasks.

Amazon

Aug 2023 – Dec 2023

Software Engineer (Intern)

Remote

- Implemented a K-means clustering module in Java for the [OpenSearch ml-commons](#) repository, contributing to distributed ML functionality and increasing unit test coverage from 66% to 78%.
- Diagnosed and resolved production data pipeline failures affecting 1M+ users, improving system reliability and reducing query latency in large-scale ML search infrastructure.

INVITED TALKS

Lawrence Berkeley National Laboratory

Mar 2026

ALPHANSO: An Open-Source Framework for (α ,n) Reactions.

Lawrence Livermore National Laboratory

Mar 2026

ALPHANSO: High-Fidelity (α ,n) Neutron Source Modeling.

TECHNICAL SKILLS

Programming: R, Python, C++, Java, MATLAB, Julia, SQL, Mathematica, $\LaTeX$.

Scientific Computing: NumPy, SciPy, Monte Carlo methods, randomized algorithms.

ML/AI: PyTorch, JAX, TensorFlow.